

AC-MOT: Adaptive Detector Operating-Point Control for Real-Time Aerial Multi-Object Tracking

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Unmanned-aerial-vehicle multi-object tracking must balance observation quality and latency while density, target scale, illumination, and blur vary within a video. We present AC-MOT, a training-free detector-control layer that changes confidence threshold, non-maximum-suppression threshold, and input resolution online while preserving the detector-tracker separation. The paper traces the method from the original five-cue scene-complexity controller through validation-calibrated historical profiles to a frozen modern policy. On the documented second read of seven VisDrone2019-MOT validation sequences, the frozen v3 policy improved HOTA over the pre-declared high-resolution baseline by 2.94 points with ByteTrack and 1.33 points with OATrack; paired sequence bootstrap intervals excluded zero. End-to-end Tesla T4 measurements were 13.45 and 15.27 frames/s, including 4.3 and 4.2 ms controller overhead. On a no-retuning UAVDT transfer, ByteTrack MOTA improved by 17.60 points; OATrack improved MOTA and identity metrics but had essentially unchanged HOTA. Attribution controls show that the evidence supports operating-point calibration and observation-set shaping relative to the declared baseline, but does not isolate a consistent additional quality contribution from the precise scene-conditioned switching schedule.

1 Introduction

Multi-object tracking from an unmanned aerial vehicle is an observation problem as much as an association problem. Density, target scale, occlusion, camera motion, illumination, and blur change within a sequence, so a tracker receives an observation stream whose quality and cost are both time-varying [1, 2]. A fixed confidence floor can remove weak but useful boxes in a sparse scene and retain clutter in a dense one; NMS can merge nearby targets; and fixed resolution spends the same detector budget on easy and difficult frames. These choices affect FP, FN, IDS, HOTA, MOTA, and throughput together.

AC-MOT was introduced as a training-free runtime layer that computes cheap scene cues, forms a Scene Complexity Index (SCI), and changes detector confidence, NMS, and input resolution while keeping detector and tracker separate. Its scientific value is the control interface: the detector operating point becomes an auditable runtime variable rather than an immutable deployment setting.

This paper follows the evolution from original AC-MOT through component ablations, validation-calibrated profiles V1 and V2, and the stricter modern policy frozen as v3. Historical evidence asks whether detector-side controls change

tracking behaviour and whether distinct calibrated profiles transfer. The modern protocol asks whether a frozen operating point policy improves a declared high-resolution baseline, whether the direction survives a second tracker host, what quality-throughput trade-off is measured on a Tesla T4, and whether exact scene-conditioned timing adds evidence beyond a useful mixture of operating points.

The contributions are bounded and evidence-oriented:

1. We reconstruct original AC-MOT, including its five cues, historical SCI, temporal smoothing, Smart Calibrator, and three detector-side controls.
2. We document the V1/V2 evolution as distinct quality, identity, and runtime profiles, preserving the component trend and historical trade-offs.
3. We specify frozen v3 at implementation level: the crowd-selected SCI, surrounding state logic, thresholds, recovery probe, hysteresis, dwell, causal information flow, and action map.
4. We evaluate ByteTrack and a controlled OATrack host with aggregate metrics and paired 5000-resample sequence bootstrap intervals.
5. We measure end-to-end Tesla T4 latency and report detector, tracker, and controller components without calling a sub-30-FPS system 30-FPS real time.
6. We report no-retuning UAVDT transfer and matched-static and shuffled- timing controls to separate baseline-relative operating-point evidence from an unproven timing-specific causal benefit.

The supported claim is precise. AC-MOT is an auditable training-free mechanism for shaping the detector observation stream and computational operating point of aerial MOT. The frozen evidence supports positive changes relative to the declared baseline and strong ByteTrack transfer, with a smaller mixed effect on OATrack. It does not establish universal improvement, official leaderboard equivalence, or that exact scene-conditioned timing is necessary for the observed accuracy gains. Historical and modern protocols are therefore kept separate throughout the paper; extended historical and audit material is provided in Supplementary Information.

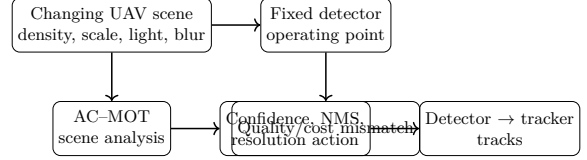


Fig. 1 Motivation for AC-MOT: a fixed detector operating point spends the same observation and compute budget across changing aerial scenes, whereas AC-MOT exposes an auditable scene-conditioned action.

2 Related work

2.1 Tracking by detection and aerial MOT

SORT established a compact online pipeline based on motion prediction and assignment, while DeepSORT added appearance information to reduce identity fragmentation [3, 4]. ByteTrack changed the detector-tracker interface by associating high-confidence detections first and then using lower-confidence boxes to recover objects that would otherwise be lost [5]. Other modern trackers further refine motion and appearance association. These trackers primarily improve the interpretation of the boxes they receive; AC-MOT asks the complementary upstream question of which observations, at which resolution and suppression policy, should reach the tracker.

VisDrone and UAVDT capture the small targets, changing target count, camera motion, view-point variation, and clutter that make aerial MOT difficult [1, 2]. In this paper they have separated roles: historical VisDrone and UAVDT evidence describes earlier profiles, while a modern frozen protocol tests v3. OATrack is a second tracker host rather than a claim of superiority over its published system [6]; its frozen repository implementation and tracker-side `min_conf=0.40` are kept distinct from the detector-side v3 confidence floor.

2.2 Detector operating points and adaptive inference

Confidence floors, NMS overlap thresholds, and input resolution are often treated as implementation settings. In aerial scenes they are coupled scientific variables: resolution changes small-object evidence and cost, confidence changes recall and

clutter, and NMS changes whether nearby targets remain separate. AC-MOT does not replace detector architecture research; it makes these deployment choices explicit and lets them respond to scene evidence without changing detector weights.

Dynamic inference systems such as Chameleon show that conditional computation can adapt inference to content and resource constraints [7]. AC-MOT is intentionally lighter: its state uses scalar image statistics and prior tracker boxes, its action set is finite, and its state machine is auditable line by line. It therefore tests an operating-point intervention at the detector-tracker boundary rather than a learned scheduler.

2.3 Evaluation and gap

MOTA emphasizes detection and identity errors, IDF1 emphasizes identity precision and recall, and HOTA balances detection and association [8]. Their different responses are useful here because selectivity may reduce FP and IDS while increasing FN. Sequence-level paired bootstrap resampling preserves within-sequence dependence while quantifying system differences [9].

The gap is consequently specific: rigorous evaluation of a training-free online controller that changes detector resolution, confidence, and NMS in aerial MOT while separating historical development from a frozen modern protocol, testing two tracker hosts, measuring runtime, transferring without retuning, and checking whether exact scene timing explains the gain. AC-MOT addresses this gap empirically and does not claim to invent adaptive inference or a new tracker.

3 Original AC-MOT

3.1 Pipeline and information flow

The original controller is placed between the image stream and an otherwise fixed detector-tracker pair. At an analysis step it receives inexpensive image statistics and information about the boxes retained by the preceding tracking step. It computes a scene state, maps that state to detector confidence, NMS, and resolution, executes the detector, and passes the resulting boxes to the tracker. The tracker output is stored for a later analysis step. This ordering is important:

the controller does not use ground truth, future frames, or the tracker output produced by the detector action it is currently choosing.

3.2 Five scene cues

Let N_t denote the number of boxes available from the preceding frame and let $a_{i,t}$ denote their source-image areas. The crowd cue is

$$C_t = \min\left(\frac{N_t}{30}, 1\right). \quad (1)$$

The tiny-object cue is the fraction of available boxes below the historical 32×32 pixel area threshold,

$$T_t = \begin{cases} N_t^{-1} \sum_i \mathbf{1}[a_{i,t} < 1024], & N_t > 0, \\ 0, & N_t = 0. \end{cases} \quad (2)$$

The edge cue is the Canny edge fraction normalized by the historical edge norm of 0.14 and clipped to one. The night cue is the indicator that mean grayscale brightness is below 80. The blur cue is the indicator that Laplacian variance is below 180. These cues were selected because they are cheap relative to detector inference and correspond to different failure modes: density, target scale, image structure, illumination, and sharpness.

The historical raw SCI is

$$S_t = 0.30C_t + 0.30T_t + 0.20E_t + 0.10N_t + 0.05B_t. \quad (3)$$

The coefficients sum to 0.95, not one. This is an intentional property of the historical implementation and is retained in the description rather than silently renormalized. Analyses are performed every ten frames and the controller uses the mean of the most recent seven analysis values. Between analysis steps the operating point is held.

3.3 Smart Calibrator

The original confidence mapping is a decreasing function of scene complexity. Before clipping and the documented scene correction, it is

$$c_t = 0.245 - 0.050S_t. \quad (4)$$

For a scene classified as crowded, tiny, or night, the code applies a -0.012 correction and clips

the result to $[0.19, 0.28]$. The correction preserves more low-score observations when the scene cue indicates that a simple scalar index may underestimate difficulty.

The NMS mapping is similarly

$$\eta_t = 0.490 - 0.050S_t. \quad (5)$$

Blur applies a -0.012 correction, and the result is clipped to $[0.40, 0.52]$. This adjustment changes how aggressively overlapping candidates are suppressed and is therefore detector-side, not a change to the tracker’s association gate.

Resolution is selected by a discrete rule. The high-complexity action uses 832 pixels when $S_t > 0.60$ or $T_t > 0.50$. The medium action uses 736 pixels when $S_t > 0.35$ or the scene label is crowded or tiny. The remaining state uses 640 pixels. The three controls are therefore adapted together, but they are not assumed to have identical monotonic effects on tracking quality.

The execution order is compactly: initialize the seven-entry SCI history and previous tracker boxes; at the first frame and every tenth frame compute the five cues from the current image and prior boxes, update the moving mean, and apply the Smart Calibrator; then run the detector at the held operating point, update the fixed tracker, and store its boxes for the next analysis step. The full historical pseudocode is retained in Supplementary Information, Sec. S3. The architecture is causal but not necessarily optimal: a held state may lag a scene transition, and a small-object action may increase association clutter. Those are empirical questions addressed by the ablation and evolution evidence.

3.4 Execution order and implementation semantics

The temporal order is part of the method. At the beginning of a frame, the controller has the previous stored tracker boxes and the current image; it computes crowd and tiny-object statistics from the previous output and edge, brightness, and blur from the current image. Only after detection and association are the new boxes stored. Using current tracker output for its own detector call would introduce a within-frame feedback loop and change the historical causal design.

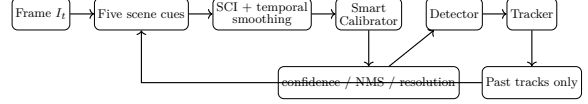


Fig. 2 Original AC-MOT information flow. Past tracker output is feedback only; the detector and tracker remain distinct, and the current decision cannot use the current frame’s future tracker output.

The image statistics use the reduced analysis image rather than a second detector. Edge density is normalized, brightness is a mean grayscale statistic, and blur is a Laplacian-variance statistic. An empty previous output gives a zero tiny ratio, while crowd saturates at one after thirty boxes, making the first decision well-defined with empty history.

Scene labels are computed alongside the scalar score: brightness determines night, Laplacian variance determines blur, the area ratio determines tiny, and density or edge conditions determine crowded. These labels permit the deterministic confidence and NMS corrections that the scalar alone would not express; no correction is learned on the final test split.

The seven-entry moving mean prevents one noisy analysis frame from changing the detector immediately, at the cost of response delay. The stride and window are historical design values, not universal optima. A reimplement should preserve the initialization, empty-box convention, stride, window, scene thresholds, corrections, clipping ranges, and resolution order; implementation details and the full trace are in Supplementary Information, Sec. S3.

4 AC-MOT evolution

The historical component ablation is the first evidence that the original controller changed the tracking problem in measurable ways. The static component configuration (OLD-A0) obtained MOTA 17.633, HOTA 29.837, IDF1 30.892, and 283 IDS. Tuning ByteTrack alone raised HOTA to 30.864. Adding adaptive confidence and NMS produced HOTA 31.384, while adaptive resolution alone reached 32.446. The full historical combination (OLD-A3) reached MOTA 18.165, HOTA 33.064, IDF1 36.296, and 271 IDS. The progression does not prove that resolution is universally

dominant, but it shows why a detector-side operating point deserves independent study: changes made before association altered both detection quality and identity outcomes.

Resolution is particularly informative because it changes the evidence available for small objects and the cost paid for every detector call. Confidence and NMS can alter the number and overlap structure of boxes at a fixed computational scale; resolution changes both evidence and computation. The component results therefore motivated a quality-oriented calibration rather than another hand adjustment of the original coefficients.

V1 replaced manually chosen controller parameters with validation-driven optimization. Its parameters were selected on the calibration data and then frozen as Trial 24 before the historical held-out evaluation. V1 should be understood as a quality-oriented profile: its purpose was to maximize the selected tracking quality objective under the historical protocol, not to establish a universal policy for every tracker. On the historical VisDrone comparison, the baseline obtained HOTA 28.430, MOTA 19.729, IDF1 32.724, IDS 1235, and 36.528 FPS. V1 obtained HOTA 33.835, MOTA 26.948, IDF1 41.546, IDS 1184, and 38.985 FPS. Its paired bootstrap changes versus baseline were +5.4051 HOTA [4.1273,6.9022], +7.2195 MOTA [5.4362,9.2934], and +8.8220 IDF1 [6.9005,11.0537]. IDS was not significant in that archived analysis. These values establish a historical quality profile with uncertainty, not a direct numerical comparison with v3.

V2 was created for a different reason. A system with a strong aggregate quality score may still produce identity fragmentation or excessive false positives, and a deployment may value a different quality-speed point. Trial 22 therefore explored a multi-objective MOTA/IDS profile. It obtained HOTA 31.218, MOTA 23.792, IDF1 37.870, IDS 919, and 46.024 FPS on the historical comparison. V2 reduced identity switches and increased speed relative to the historical baseline, but its HOTA was below V1. It is consequently a deliberate identity/false-positive/runtime trade-off, not a global replacement for the quality-oriented V1.

The historical UAVDT transfer tests the same frozen profiles without retuning. Baseline HOTA/MOTA/IDF1/IDS were 24.085/13.841/27.887/558; V1 obtained

Table 1 Compact historical evolution summary. These rows belong to the legacy protocol and are not pooled with modern v3 results; full ablation, bootstrap, and UAVDT rows are in Supplementary Information, Sec. S1.

Profile		HOTA	MOTA	IDS	FPS
Original	full	33.064	18.165	271	37.962
AC-MOT					
V1	quality	33.835	26.948	1184	38.985
profile					
V2	iden- tity/speed profile	31.218	23.792	919	46.024

28.390/17.399/34.453/321; V2 obtained 26.930/16.118/32.014/308. The transfer supports the portability of frozen operating profiles, while the different V1/V2 patterns reinforce that tracker quality, identity stability, and throughput are not reducible to one scalar objective.

The evolution therefore has a clear logic. Original AC-MOT established a lightweight five-cue controller. Ablation showed that detector controls affected downstream tracking. V1 calibrated the quality profile. V2 explored a different deployment trade-off. The modern redesign then tightened provenance, split development from evaluation, tested two tracker hosts, and added runtime and attribution controls because the historical results alone could not identify whether the improvement came from dynamic timing, operating-point calibration, tracker interaction, or some combination.

This chronology also prevents anachronistic interpretation of the final action map. The historical controller’s 640/736/832 rule was a hand-designed scene-to-resolution mapping coupled to confidence and NMS formulas. V1 and V2 changed the selection problem rather than merely changing a threshold. Modern v3 then used a different detector and action set. The fact that v3 has LOW=1536 and HIGH=1088 is not an error inherited from the historical design; it is a frozen result of a modern calibration search whose action names describe state labels rather than resolution rank.

5 Modern frozen AC–MOT v3

5.1 Motivation and development boundary

The modern redesign was needed because the historical controller had been evaluated through several intertwined choices: detector, tracker, operating grid, evaluation protocol, and selection objective. A stronger public YOLO11m checkpoint made detector-side sensitivity a central question, while earlier calibration searches showed that an adaptive policy could fail to beat a matched static point. The modern study therefore froze a detector checkpoint, declared a calibration split, used sequence-level cross-validation, retained negative search results, and evaluated two tracker hosts. It also added bootstrap uncertainty, end-to-end timing, frozen UAVDT transfer, and attribution controls.

Development used eight VisDrone2019-MOT calibration sequences comprising 3282 frames. Four folds held out sequence pairs during calibration; the final seven-sequence modern VisDrone evaluation was not used to choose the policy. The detector is `dronefreak/visdrone-yolo11m`, trained on VisDrone-DET at 640 pixels, with checkpoint hash `c6f98247e7c731a567aaf37ca7b808beff588b687623e028b25a2e1d846e96e7`. This is not the OATrack detector and is not a VisDrone-MOT-trained checkpoint.

5.2 Cue audit and modern SCI

The search first audited which scene information was useful under the new detector and protocol. Stage 3A selected the cue subset [`'crowd'`]. The result must be stated carefully. The weighted modern SCI is crowd-selected; the wider SceneLayer still computes tiny-object, edge, brightness, and blur statistics because those values participate in scene labels, the tiny guard, and the recovery and hysteresis machinery.

At analysis step j , let N_j be the number of boxes reported by the tracker before the current detector decision. The crowd measurement is

$$C_j = \min(N_j/30, 1). \quad (6)$$

The weighted v3 index is the mean of the most recent up-to-seven crowd measurements,

$$S_j = \frac{1}{|H_j|} \sum_{k \in H_j} C_k, \quad (7)$$

where H_j is the current seven-entry history. Analyses occur every ten frames. The current frame's tracker output is not available when S_j is computed; it is stored after detection and association for future decisions.

5.3 State target and surrounding guards

The raw target state is HIGH when $S_j > 0.5114928923560124$ or the tiny-object ratio exceeds 0.50. It is MEDIUM when $S_j > 0.29747709701135766$ or the scene label is crowded or tiny. Otherwise it is LOW. HIGH is a state label, not a statement that HIGH uses the largest input resolution.

The target is then processed by the state-stabilization logic. The scene label is **night** when brightness is below 80, **blur** when Laplacian variance is below 180, **tiny** when the tiny ratio exceeds 0.50, and **crowded** when crowd or edge conditions exceed their historical scene thresholds. These labels do not add weight to the selected SCI, but they can guard a state transition. The implementation tracks a decaying peak count. After at least five observed boxes, a count below 0.40 of the decaying peak increments the drop age; during up to three analysis steps after such a collapse, the recovery probe requests HIGH. The probe is a bounded response to a sudden loss of observations, not access to future frames or ground truth.

Downward transitions are held by hysteresis. A HIGH state remains HIGH when $S_j > 0.50$ or tiny exceeds 0.40; a MEDIUM state remains MEDIUM when $S_j > 0.25$. Finally, a 30-frame dwell rule prevents a new target from changing the state within 30 frames of the preceding change. The ordering is target, recovery probe, downward holds, dwell, and state update. The resulting machine is causal and auditable, but it is not asserted to be an optimal controller.

Algorithm 1 Frozen v3 causal controller

```

1: initialize level LOW, empty seven-entry history, empty previous boxes, peak count 0
2: for each frame  $t$  do
3:   if  $t = 1$  or  $t \bmod 10 = 1$  then
4:     compute image statistics and crowd/tiny values using boxes from frames  $< t$ 
5:     append  $C_j$  and set  $S_j$  to the history mean
6:     set target HIGH, MEDIUM, or LOW using  $S_j$ , tiny, and scene label
7:     update decaying peak; if collapse gate passes, apply up to three-step HIGH probe
8:     apply downward holds, then reject changes inside the 30-frame dwell interval
9:     update persistent level
10:  end if
11:  select frozen action for the persistent level; run detector and confidence filter
12:  update the fixed tracker and store its boxes for future frames
13: end for

```

5.4 Frozen action map

The frozen actions are LOW (1536, 0.70, 0.10), MEDIUM (1280, 0.45, 0.40), and HIGH (1088, 0.60, 0.40), where each tuple is resolution, NMS IoU, and detector confidence floor. The apparent non-monotonicity is intentional. State names describe scene difficulty, whereas the action was selected from the calibration search and need not increase resolution with state rank. Reporting the names as resolution ranks would reverse the meaning of the frozen method.

This algorithm makes the interface explicit. AC-MOT controls the detector operating point; ByteTrack or OATrack remains responsible for association. The detector-side confidence floor is not OATrack’s tracker-side `min_conf=0.40`. No detector scores enter the scene layer, no online labels are used, and no parameter is changed after the freeze.

5.5 Causal execution and implementation boundary

At an analysis frame, the current image supplies edge, brightness, and blur statistics, while tracker boxes from frames $< t$ supply crowd, tiny-object,

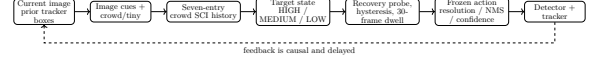


Fig. 3 Frozen v3 state machine. The weighted SCI uses the selected crowd cue, while tiny, edge, brightness, blur, recovery, hysteresis, and dwell logic remain part of the surrounding causal SceneLayer.

Table 2 Frozen modern v3 action map. State names are controller labels, not a monotonic resolution ranking.

State	Resolution	NMS IoU	Conf. floor
LOW	1536	0.70	0.10
MEDIUM	1280	0.45	0.40
HIGH	1088	0.60	0.40

and peak-count values. The seven-entry crowd history is updated, the target state is computed, and the collapse probe, hysteresis, and 30-frame dwell are applied in that order. The persistent state then selects the frozen tuple before detector execution; tracker output is stored only for later decisions. Thus the crowd-selected SCI must not be shortened to “crowd-only controller”: tiny guards the high state and recovery, brightness and blur form scene labels, edge can produce a crowded label, and temporal memory remains active. No cue reads detector scores, ground truth, or future frames. The full implementation trace is retained in Supplementary Information, Sec. S3.

LOW, MEDIUM, and HIGH are state names, not an ordered resolution scale. The frozen mapping is LOW 1536, MEDIUM 1280, and HIGH 1088; an implementation must use these tuples rather than substitute a monotonic resolution policy.

6 Experimental protocol**6.1 Historical and modern separation**

Historical results come from the legacy freeze: original AC-MOT, V1 Trial 24, V2 Trial 22, historical VisDrone test-dev, and historical UAVDT transfer. Modern results come from tag `acmot-v3-exp-2026-10-04-freeze` and its archived experiment package. The protocols are not pooled because their detectors, operating grids, evaluation details, and held-out roles differ.

6.2 Modern detector, splits, and transfer

The modern detector is `dronefreak/visdrone-yolo11m` trained on VisDrone2019-DET at 640 pixels; the checkpoint SHA256 is `c6f98247e7c731a567aaf37ca7b808beff588b687623e028b25a2e1d846e96e7`. Development uses eight VisDrone2019-MOT calibration sequences and four sequence-level folds. The modern VisDrone read uses seven sequences and 71,830 ground-truth boxes. It is the project’s class-agnostic protocol, not an official VisDrone leaderboard evaluation, and it is disclosed as a second validation read because a static control had already read the sequences.

UAVDT transfer uses 20 sequences, 16,592 frames, and 49,776 detector forward passes from three action-specific cache pairs. The frozen policy is applied without recalibration or retuning, once per tracker host, so no bootstrap or cross-validation claim is attached.

6.3 Trackers, baseline, and metrics

ByteTrack uses the frozen archive defaults. OATrack is the repository’s frozen reimplementa-tion with tracker-side `min_conf=0.40`; this is separate from the detector-side v3 confidence floor. The declared modern baseline is `r1536_n70`: resolution 1536, NMS IoU 0.70, and no detector filter above the cache floor 0.01. V3 uses the frozen action map and thresholds; no detector, tracker, or operating choice changes after the freeze.

We report HOTA, MOTA, IDF1, IDS, FP, FN, precision, and recall. HOTA balances detection and association, MOTA exposes the combined FP/FN/IDS effect, and IDF1 measures identity precision and recall. Baseline and v3 are paired by sequence. The archived modern bootstrap uses 5000 paired resamples with seed 42; it is not applied to the single-run UAVDT transfer.

6.4 Runtime and evaluation handling

V3 runtime is measured on a Tesla T4 over 200 frames with causal scene analysis, detector execution at the selected resolution, and tracker update. Mean latency, P95 latency, FPS, detector time, tracker time, and controller overhead are recorded. The baseline harness does not include

image-read work identically, so the comparison is reported with that caveat and is not called a perfectly matched speedup. The analysis does not claim a 30-FPS deployment target.

Detector outputs are cached at the required resolution/NMS combinations for reproducibility. The project maps classes to its documented class-agnostic vehicle/tracking protocol before TrackEval-style scoring; these values must therefore not be described as official leaderboard metrics. The archive records the detector hash, controller thresholds, action map, tracker descriptions, calibration folds, forward-pass count, and source checksums. It also records that frozen Trial 7 is not the literal argmax of the documented selection rule under either eligibility interpretation; the paper reports this deviation and does not rerun the search.

7 Results

7.1 Historical reference

The historical chain provides context without being pooled with v3. Original AC-MOT reached HOTA 33.064 in the component ablation; V1 reached HOTA 33.835 and MOTA 26.948; V2 reached HOTA 31.218 while reducing IDS to 919 and raising FPS to 46.024. V1 is therefore a quality-oriented profile, whereas V2 is an identity/runtime trade-off. The full archived rows and uncertainty are in Supplementary Information, Sec. S1.

7.2 Modern VisDrone

Table 3 reports the aggregate results on the documented seven-sequence second validation read. For ByteTrack, v3 changes HOTA from 41.586 to 44.523, MOTA from 9.876 to 27.871, IDF1 from 48.569 to 53.929, and IDS from 1341 to 1016. FP falls from 39274 to 24892, while FN rises from 24121 to 25902. This is a selective observation stream: clutter and identity- confusing candidates decrease at a recall cost.

For OATrack, HOTA changes from 46.770 to 48.095, MOTA from 28.165 to 34.483, IDF1 from 57.107 to 59.364, and IDS from 773 to 604. FP falls from 25394 to 21400 and FN changes from 25432 to 25057. The smaller increment is consistent

with a stronger baseline and different association response, not evidence that the controller is ineffective. The two hosts are therefore reported separately rather than pooled into one AC-MOT gain.

The paired bootstrap in Table 4 supports the main positive differences. ByteTrack HOTA, MOTA, IDF1, and IDS changes are +2.937 [2.356,3.375], +17.995 [13.541,23.282], +5.360 [4.490,6.178], and -325 [-498,-164]. OATrack changes are +1.325 [0.200,2.264], +6.318 [4.741,7.879], +2.258 [0.548,3.474], and -169 [-292,-42]. The ByteTrack FN increase has CI [288,3594.075], whereas OATrack FN has CI [-1462.575,670]; the evidence is consequently a quality trade-off rather than an unqualified recall improvement.

The state occupancy is reported in Supplementary Information, Sec. S2. It confirms that both hosts use multiple actions and that their trajectories differ, but occupancy alone does not establish timing causality.

8 Runtime and real-time operating points

The runtime experiment measures scene analysis, state update, detector execution at the selected resolution, confidence filtering, and tracker update on a Tesla T4 over 200 frames. This distinguishes a controller that shapes compute over time from a uniformly larger detector budget.

For ByteTrack, v3 mean latency is 74.3238 ms, P95 latency 100.6847 ms, and throughput 13.4546 FPS; detector, tracker, and controller components are 54.5688, 7.9712, and 4.2562 ms. For OATrack, mean latency is 65.4748 ms, P95 85.8969 ms, and throughput 15.2731 FPS; the components are 52.6399, 1.8355, and 4.2010 ms. The detector remains dominant and the controller is small but nonzero, at roughly four milliseconds per frame.

The high-resolution baseline harness reports 139.93 ms and 7.15 FPS for ByteTrack and 128.85 ms and 7.76 FPS for OATrack. Image-read work is not included identically, so this is not a perfectly component-matched speedup. The safe interpretation is reduced measured detector time from the selected actions plus explicit controller overhead.

Neither 13.4546 nor 15.2731 FPS is claimed as 30-FPS real-time operation; the P95 values also do not establish a hard per-frame deadline.

The tracker component differs substantially between hosts while detector times are similar, supporting separate tracker-dependent reporting. A future cost-matched study would need equal mean detector work and identical image-read instrumentation to isolate scheduling from average compute.

9 Frozen UAVDT transfer

The modern transfer applies frozen v3 to UAVDT without retuning thresholds, weights, action mapping, or tracker configuration. It contains 20 sequences, 16,592 frames, and 49,776 detector forward passes from three action-specific cache pairs. Each host is run once without bootstrap or cross-validation.

For ByteTrack, baseline/v3 values are HOTA 45.923/47.004, MOTA -1.573/16.028, IDF1 58.400/61.767, IDS 601/486, FP 279616/212077, and FN 66050/73702. The changes are HOTA +1.081, MOTA +17.601, IDF1 +3.367, IDS -115, and FP -67539; the false-positive reduction accompanies a false-negative increase of 7652. This supports a strong MOTA/precision direction for ByteTrack, not an uncertainty-certified universal improvement.

For OATrack, baseline/v3 values are HOTA 47.728/47.640, MOTA 19.607/22.873, IDF1 62.101/62.931, IDS 517/486, and FP 197937/183669. MOTA, IDF1, IDS, and FP improve, but HOTA changes by -0.087. This mixed transfer is why the result is reported per tracker host. The class-agnostic adapter is not an official UAVDT leaderboard protocol, and the VisDrone-trained detector does not make this a detector-domain adaptation result.

10 Attribution and discussion

The experiments establish that detector operating point affects downstream MOT, that v3 changes the declared high-resolution baseline on two tracker hosts, and that operating-point choices shape measured compute. The remaining question is whether the observed quality difference is caused specifically by the exact scene-conditioned switching schedule.

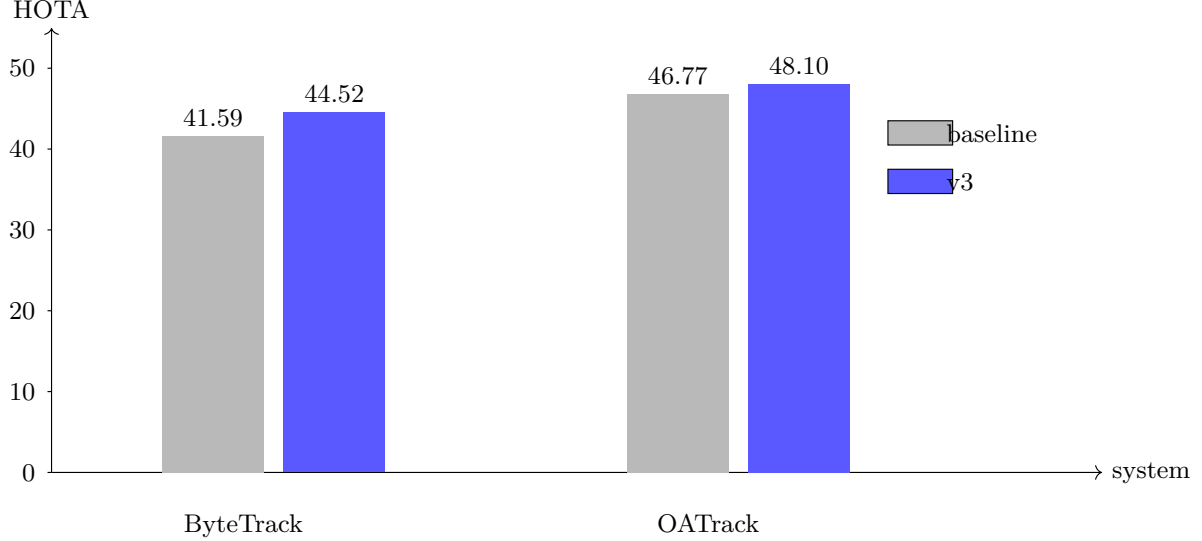


Fig. 4 Modern VisDrone results on the documented seven-sequence second validation read. The adaptive policy is compared with the predeclared high-resolution baseline separately for ByteTrack and OATrack; the custom class-agnostic protocol and read-order caveat apply.

Table 3 Modern VisDrone aggregate results on seven sequences and 71,830 ground-truth boxes. V3 is a documented second validation read.

Host	Policy	HOTA	MOTA	IDF1	IDS	FP	FN	Precision	Recall
ByteTrack	baseline	41.586	9.876	48.569	1341	39274	24121	54.849	66.419
ByteTrack	v3	44.523	27.871	53.929	1016	24892	25902	64.852	63.940
OATrack	baseline	46.770	28.165	57.107	773	25394	25432	64.628	64.594
OATrack	v3	48.095	34.483	59.364	604	21400	25057	68.609	65.116

Table 4 Modern paired bootstrap changes versus the predeclared baseline. The intervals use 5000 paired sequence resamples with seed 42.

Host	Metric	Difference	95% CI
ByteTrack	HOTA	+2.937	[+2.356, +3.375]
ByteTrack	MOTA	+17.995	[+13.541, +23.282]
ByteTrack	IDF1	+5.360	[+4.490, +6.178]
ByteTrack	IDS	-325	[-498, -164]
OATrack	HOTA	+1.325	[+0.200, +2.264]
OATrack	MOTA	+6.318	[+4.741, +7.879]
OATrack	IDF1	+2.258	[+0.548, +3.474]
OATrack	IDS	-169	[-292, -42]

The matched static control uses resolution 1088, NMS IoU 0.45, and detector confidence 0.40. Its HOTA is approximately 44.530 for ByteTrack and 48.269 for OATrack, close to v3 values 44.523 and 48.095. The comparison against `r1536_n70` therefore supports moving away from the declared

baseline, but does not isolate an additional accuracy benefit from changing the action frame by frame.

The timing control has the same implication: development mean Δ HOTA is 2.446957 for the causal schedule and 2.617228 for a same-action-distribution shuffled schedule. This does not show that random timing is globally superior; both are calibration statistics from one frozen control. It does show that the present evidence cannot claim exact scene-conditioned timing as necessary for the baseline-relative gain.

The controller nevertheless has systems value when much of the accuracy difference is explained by operating-point mixture. It provides online compute allocation, reacts to changing scene statistics, and exposes a finite action interface without retraining detector weights. A dynamic

policy may therefore be preferable under an average detector-time budget even when a matched static point has similar HOTA. This is a deployment interpretation, not a new causal claim.

Tracker dependence supports keeping the hosts separate. ByteTrack starts from a weaker modern baseline and shows a larger VisDrone change; OATrack starts stronger and has a smaller HOTA increment and essentially flat UAVDT HOTA transfer. The detector-side policy changes the observations available to each association algorithm, so a pooled “AC–MOT gain” would be misleading.

The negative searches also bound the conclusion. Resolution-only, NMS-only, cue-subset, and earlier joint formulations did not pass the predeclared matched-static robustness gate. The final candidate was selected against `r1536_n70`, not because the archive established that scene timing beat every fixed point. The selection audit records that Trial 7 was not the literal argmax under either eligibility interpretation; the freeze is preserved and the deviation is reported as provenance.

The strongest supported conclusion is that AC–MOT is an auditable, training-free mechanism for shaping the detector observation stream and computational operating point of aerial MOT. It can improve a declared baseline and transfer with tracker-dependent outcomes. It does not yet establish that exact causal switching adds a separate, consistent accuracy benefit beyond operating-point calibration and action-mixture effects. Cost-matched randomized policies, more detectors and trackers, and a predeclared dynamic-scheduling objective are the appropriate future tests.

11 Limitations

The modern seven-sequence VisDrone result is a second validation read: a static control had already read those sequences before v3. It is therefore a frozen, documented comparison, not an untouched held-out confirmation; the historical test-dev experiment is the cleaner historical held-out evidence.

The freeze records that Trial 7 is not the literal argmax of the documented selection rule under either archived eligibility interpretation. This provenance limitation is reported without

rerunning the search. Matched-static and shuffled-timing controls further weaken a claim that exact switching timing causes the accuracy gain. The project class-agnostic evaluator is not an official VisDrone leaderboard protocol.

Only one modern detector checkpoint is used: a third-party VisDrone-DET-trained YOLO11m model rather than a VisDrone-MOT-trained or published OATrack detector. Two tracker hosts demonstrate interaction but not tracker independence, and the OATrack implementation is a controlled repository reimplement rather than a direct published-system comparison.

UAVDT transfer is a single frozen-policy run per host without cross-validation or bootstrap. Its ByteTrack result is strongly positive in MOTA and FP, while OATrack HOTA is essentially flat. T4 timing is measured, but image-read work is not instrumented identically in baseline and v3 harnesses. Finally, throughput is below 30 FPS. These limits constrain generalization and attribution while leaving a reproducible detector-operating-point study with measured quality, cost, and transfer behaviour.

12 Conclusion

This paper followed AC–MOT as a scientific evolution rather than as one isolated controller. Original AC–MOT introduced a five-cue SCI, temporal smoothing, and a Smart Calibrator that adapted detector confidence, NMS, and resolution without retraining. Historical component ablations showed that these detector-side choices changed downstream tracking. V1 then provided a validation-calibrated quality profile, while V2 explored a deliberate identity, false-positive, and runtime trade-off. Historical VisDrone and UAVDT results demonstrate why the profiles must be treated as distinct operating points rather than collapsed into one universal winner.

The modern frozen v3 policy tightened the protocol. It selected a crowd-weighted SCI on a calibration split, retained explicit causal state machinery with recovery, hysteresis, and dwell logic, and mapped LOW, MEDIUM, and HIGH states to three frozen detector actions. Two tracker hosts, paired bootstrap intervals, T4 timing, and no-retuning UAVDT transfer made the quality–compute interaction measurable. The modern results support positive changes over the

declared `r1536_n70` baseline on both tracker hosts and a strong ByteTrack transfer, with mixed OATrack transfer.

The final scientific conclusion is narrower than a claim that switching always helps. Matched-static results are close to `v3` and shuffled timing is at least as good on calibration, so the current evidence does not establish that exact scene-conditioned switching timing is necessary for the observed accuracy gains. It supports training-free detector operating-point control as a useful mechanism for shaping tracking quality and compute, while leaving causal timing benefit, universal tracker independence, and broader detector generalization for explicitly designed future experiments.

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Consent for publication. Not applicable.

Data and code availability. The frozen controller specification, result records, evidence map, staged sections, scripts, and build package are included in the project repository. Dataset redistribution remains subject to the original benchmark licenses and access terms.

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References

- [1] Zhu, P., Wen, L., Du, D., Bian, X., Fan, H., Hu, Q., Ling, H.: Detection and Tracking Meet Drones Challenge. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **44**(11), 7380–7399 (2022) <https://doi.org/10.1109/tpami.2021.3119563>
- [2] Du, D., Qi, Y., Yu, H., Yang, Y., Duan, K., Li, G., Zhang, W., Huang, Q., Tian, Q.: The Unmanned Aerial Vehicle Benchmark: Object Detection and Tracking. In: *Computer Vision – ECCV 2018. Lecture Notes in Computer Science*, pp. 375–391. Springer, ??? (2018). https://doi.org/10.1007/978-3-030-01249-6_23
- [3] Bewley, A., Ge, Z., Ott, L., Ramos, F., Upcroft, B.: Simple online and realtime tracking. In: *2016 IEEE International Conference on Image Processing (ICIP)*, pp. 3464–3468. IEEE, ??? (2016). <https://doi.org/10.1109/icip.2016.7533003>
- [4] Wojke, N., Bewley, A., Paulus, D.: Simple online and realtime tracking with a deep association metric. In: *2017 IEEE International Conference on Image Processing (ICIP)*, pp. 3645–3649. IEEE, ??? (2017). <https://doi.org/10.1109/icip.2017.8296962>
- [5] Zhang, Y., Sun, P., Jiang, Y., Yu, D., Weng, F., Yuan, Z., Luo, P., Liu, W., Wang, X.: ByteTrack: Multi-object Tracking by Associating Every Detection Box. In: *Computer Vision – ECCV 2022. Lecture Notes in Computer Science*, pp. 1–21. Springer, ??? (2022). https://doi.org/10.1007/978-3-031-20047-2_1
- [6] Qi, L., Wang, L., Jiang, S.: OATrack: A Quality-Gated Progressive Association Framework with a YOLO Detection Cache for UAV Small-Object Multi-Object Tracking. *Sensors* **26**(16), 5222 (2026) <https://doi.org/10.3390/s26165222>
- [7] Jiang, J., Ananthanarayanan, G., Bodik, P., Sen, S., Stoica, I.: Chameleon: Scalable Adaptation of Video Analytics. In: *Proceedings of the 2018 Conference of the ACM Special Interest Group on Data Communication*, pp. 253–266. ACM, ??? (2018). <https://doi.org/10.1145/3230543.3230574>
- [8] Luiten, J., Ošep, A., Dendorfer, P., Torr, P., Geiger, A., Leal-Taixé, L., Leibe, B.: HOTA: A Higher Order Metric for Evaluating Multi-object Tracking. *International Journal of Computer Vision* **129**(2), 548–578 (2021) <https://doi.org/10.1007/s11263-020-01375-2>
- [9] Efron, B., Tibshirani, R.J.: *An Introduction to the Bootstrap*. Springer, ??? (1993). <https://doi.org/10.1007/978-1-4899-4541-9>